

Counts with multiplicity at general orders

Companion note to “A binomial count of meromorphic travelling waves”

Alexander Migita

Abstract

The paper proves its counts when $2p - 1$ is prime, by reduction modulo that prime. This note contains what is known at the other orders. A weighted form of Bézout’s theorem gives the upper bound $\binom{2p-1}{p-1}$ at every order, and equality, with pairs counted with multiplicity, whenever a rigidity condition holds. The same method gives the counts of pulses, of the pairs of the two pure subfamilies and of elliptic pairs. The note also proves rigidity when $2p - 1$ is a prime power and records the computations that establish it for $p \leq 573$, and distinctness of all pairs for $p \leq 12$. The text is that of the first complete version of the paper, where these results were part of the main line. The last section keeps two algebraic arguments of an earlier version which the paper now presents in other terms: the general lemma on reduction modulo a prime, and a proof by flatness that the number of elliptic pairs is conserved. It also contains the full proof of the conservation-of-number lemma, of which the paper gives a sketch.

1 Notation

References of the form “Theorem P 3.4” are to the paper. We use its notation throughout. The profile equation is $P(D)v + v^2/2 = 0$ with P monic of degree p (equation (1.1) of the paper). A rational-exponential pair has the profile $v_A = s_p A(D)Q$, where $Q = 1/(1 + e^{-z})$ and $A(\rho) = \rho^d + \alpha_1 \rho^{d-1} + \cdots + \alpha_d$, $d = p - 1$. The discrete convolution is $S_A = A \star A$, and the pair exists exactly when $A \mid S_A$ (Proposition P 3.1). Writing $S_A = AC_A + R_A$, $R_A = \sum_{r < d} R_r \rho^r$, the matching equations are $R_0 = \cdots = R_{d-1} = 0$, and $\mathcal{X}_p$ is the affine scheme that they define. A solution is *simple* if the Jacobian matrix of the R_r is invertible there. When $\mathcal{X}_p$ is finite, its *length* is the sum of the multiplicities of its points, and $\mathcal{X}_p$ is *reduced* when every point is simple, so that the length is the number of points. We put

$$N_p = \binom{2p-1}{p-1}, \quad S_p = \binom{2p-2}{p-2}, \quad F_p = \binom{p-1}{\lfloor (p-1)/2 \rfloor}, \quad M_p = 2S_p.$$

The variable α_j has weight j and ρ has weight one; $H_A(\rho) = \int_0^\rho A(t)A(\rho - t) dt$ is the continuous convolution, and (R_d) is the rigidity statement (3.8) of the paper: a monic A of degree d with $A \mid H_A$ is ρ^d . The reflection is $(\iota A)(\rho) = (-1)^d A(-\rho)$, with the identities of Proposition P 6.1. For a fixed lattice, $\mathcal{E}_p(g_2, g_3)$ is the elliptic matching system of Section P 4.

2 Weighted degree and the count with multiplicities

Bézout’s theorem bounds the sum of the multiplicities of the isolated solutions of d polynomial equations in d unknowns by the product of their degrees. Equality holds when the system has no solutions at infinity, that is, when the highest-degree parts of the equations have no common zero

other than the origin. The ordinary degrees of the matching equations give a poor bound. The right measure of their size is a weighted degree that reflects how the α_j scale. Give α_j weight j , and ρ weight one. The remainder equations have highest weighted degrees

$$2d + 1 - r \quad (0 \leq r < d), \quad \text{namely } d + 2, \dots, 2d + 1. \quad (2.1)$$

Their highest weighted parts are the coefficients of $\text{rem}_A H_A$, where

$$H_A(\rho) = \int_0^\rho A(t)A(\rho - t) dt. \quad (2.2)$$

Indeed, if $A(\rho) = \sum a_i \rho^i$, then

$$H_A(\rho) = \sum_{i,j=0}^d \frac{i!j!}{(i+j+1)!} a_i a_j \rho^{i+j+1}.$$

These are precisely the highest weighted terms of the power sums defining S_A , and monic division preserves the highest weighted part. Indeed, A is weighted-homogeneous of weight d , and each step of long division subtracts a multiple of A of the same weight as the term being removed. In words, the discrete convolution S_A is a lower-order perturbation of the continuous convolution H_A , and the behaviour of the matching equations at infinity is governed by the continuous problem $A \mid H_A$.

For this degree, denote by (R_d) the condition

$$A \text{ monic}, \quad \deg A = d, \quad A \mid H_A \implies A(\rho) = \rho^d. \quad (2.3)$$

It says that the continuous problem has only the trivial solution. By the preceding paragraph, it is exactly the condition that the matching equations have no solution at weighted infinity.

Theorem 2.1 (Count with algebraic multiplicities). *For every $p \geq 2$, the sum of multiplicities of the isolated points of $\mathcal{X}_p$ is at most*

$$N_p = \binom{2p-1}{p-1}.$$

If (R_{p-1}) holds, then $\mathcal{X}_p$ is finite and

$$\text{length } \mathcal{X}_p = N_p. \quad (2.4)$$

This equality does not require simplicity.

Proof. Weighted degree is turned into ordinary degree by the finite substitution

$$\alpha_j = y_j^j, \quad 1 \leq j \leq d. \quad (2.5)$$

A monomial of weight w in the α_j becomes a monomial of ordinary degree w in the y_j , so the pulled-back equations have ordinary degrees at most $d + 2, \dots, 2d + 1$, and their highest-degree parts are the pull-backs of the highest weighted parts. Bézout's inequality for isolated solutions bounds the sum of their multiplicities by $(d+2) \cdots (2d+1)$; see Mondal [6, Chapter VIII]. If (R_d) holds, the leading homogeneous forms have no nonzero common zero. The homogenized system has no point at infinity, and ordinary Bézout gives equality [2].

It remains to return from the y_j to the α_j . A solution in the α_j with all $\alpha_j \neq 0$ has $1 \cdot 2 \cdots d = d!$ preimages in the y_j , one for each choice of the roots $y_j = \alpha_j^{1/j}$, each with the original multiplicity. In general, for each solution the sum of the multiplicities of its preimages is $d!$ times its original multiplicity, including at branch points where some α_j vanish. Thus the substitution multiplies the total length by $d!$. Algebraically, $\mathbb{C}[y_1, \dots, y_d]$ is a free module over $\mathbb{C}[\alpha_1, \dots, \alpha_d]$, with basis

$$y_1^{e_1} \cdots y_d^{e_d}, \quad 0 \leq e_j < j.$$

This remains true after imposing the equations and restricting to an isolated component. Dividing the Bézout count by $d!$ gives

$$\frac{(d+2) \cdots (2d+1)}{d!} = \binom{2d+1}{d}.$$

The no-infinity condition also gives finiteness of the whole affine scheme, completing the proof. $\square$

For $p = 3$, the variable weights are 1, 2 and the equation weights are 4, 5, so the calculation is $(4 \cdot 5)/(1 \cdot 2) = 10$.

3 Pulses

The endpoints of v_A are 0 and $s_p A(0)$; the pairs with $A(0) = 0$ are the pulses.

Proposition 3.1 (Equal endpoints). *Assume (R_{p-1}) , and let $\mathcal{Z}_p = \mathcal{X}_p \cap \{A(0) = 0\}$, with its induced scheme structure. Then*

$$\text{length } \mathcal{Z}_p = S_p := \binom{2p-2}{p-2}. \quad (3.1)$$

If $\mathcal{X}_p$ is reduced, these are S_p distinct pairs and $P_A(0) \neq 0$ at each of them. In particular, under the prime-index hypothesis, exactly

$$\binom{2p-2}{p-2}$$

normalized pairs have equal zero endpoints. The remaining $\binom{2p-2}{p-1}$ have unequal endpoints. These are counts over $\mathbb{C}$.

Proof. Polynomial summation gives $S_A(0) = -A(0)^2$, so the constant remainder satisfies the polynomial identity

$$R_0 = -A(0)(A(0) + C_A(0)). \quad (3.2)$$

Set $\alpha_d = A(0) = 0$. The equation $R_0 = 0$ disappears, leaving $d-1$ equations of weights $d+2, \dots, 2d$ in variables of weights $1, \dots, d-1$. Their leading forms have no nonzero common zero by (R_d) . The weighted calculation in Theorem 2.1 gives

$$\text{length } \mathcal{Z}_p = \frac{(d+2) \cdots (2d)}{(d-1)!} = \binom{2d}{d-1}.$$

For $d = 1$ the system is empty and its length is one.

If $\mathcal{X}_p$ is finite and reduced, its coordinate algebra is a product of copies of $\mathbb{C}$; the same is true of any quotient, in particular that of $\mathcal{Z}_p$, so $\mathcal{Z}_p$ is reduced as well. At a point with $A(0) = C_A(0) = 0$, the differential of (3.2) vanishes. Such a point cannot be simple in the square matching system. Therefore $C_A(0) \neq 0$, and $P_A(0) = s_p C_A(0)/2 \neq 0$, at every point of $\mathcal{Z}_p$ when $\mathcal{X}_p$ is reduced. The prime-index assertions now follow from Theorem P 3.4 and $N_p - S_p = \binom{2p-2}{p-1}$. $\square$

4 The two pure subfamilies

Let

$$\mathcal{X}_p^{\text{sym}} = \mathcal{X}_p \cap \{\alpha_j = 0 \text{ for odd } j\}$$

be the fixed locus of ι , with its induced scheme structure, and put

$$F_p = \binom{p-1}{\lfloor (p-1)/2 \rfloor}. \quad (4.1)$$

Its first values are 1, 2, 3, 6, 10, 20 for $p = 2, \dots, 7$; Table P 1 lists the equations to which they belong.

Theorem 4.1 (Count in the two pure subfamilies). *(i) A point A of $\mathcal{X}_p$ lies in $\mathcal{X}_p^{\text{sym}}$ if and only if P_A is reflection-symmetric. Thus $\mathcal{X}_p^{\text{sym}}$ consists of the rational-exponential pairs of the purely dispersive equations when p is even, and of the purely dissipative equations when p is odd.*

(ii) For every p the isolated points of $\mathcal{X}_p^{\text{sym}}$, counted with multiplicity, number at most F_p . If (R_{p-1}) holds, then $\text{length } \mathcal{X}_p^{\text{sym}} = F_p$.

(iii) If, in addition, $\mathcal{X}_p$ is reduced, then $\mathcal{X}_p^{\text{sym}}$ consists of F_p distinct pairs, each a simple solution of the matching equations of its subfamily. If $2p-1$ is prime, they are the points whose label in (P 3.9) satisfies $L = -L$.

(iv) For even p , every pair of $\mathcal{X}_p^{\text{sym}}$ is a pulse. For odd p , every pair of $\mathcal{X}_p^{\text{sym}}$ is a front if $\mathcal{X}_p$ is reduced, and every pair with real coefficients is a front without any hypothesis.

In plain terms, the pairs that coincide with their mirror image are exactly the pairs of the purely dispersive equations, for even p , or of the purely dissipative equations, for odd p . Counted with multiplicity, the isolated ones number at most F_p at every order, and there are exactly F_p of them whenever rigidity holds, in particular for $p \leq 573$. They are F_p distinct pairs whenever all N_p pairs for that p are distinct, in particular when $2p-1$ is prime. The rational-exponential waves of a purely dispersive equation are pulses, for every even p . Those of a purely dissipative equation are fronts; this is proved whenever all N_p pairs are distinct, and for waves with real coefficients at every odd p . Elliptic waves occur in both subfamilies; the distinction between pulses and fronts does not apply to them.

Proof. (i). If $\iota A = A$, then (P 6.3) gives $P_A(\rho) = (-1)^p(P_A(-\rho) + s_p A(0))$. For even p the polynomial A is odd, so $A(0) = 0$ and P_A is even. For odd p the sum $P_A(\rho) + P_A(-\rho)$ is the constant $-s_p A(0)$. In both cases P_A is reflection-symmetric, and

$$s_p A(0) = -2P_A(0) \quad (p \text{ odd}). \quad (4.2)$$

Conversely, let $P = P_A$ be reflection-symmetric, and apply Corollary P 5.3 at the pole $i\pi$, using the period $2\pi i$. If p is even, then $v_A(-z) = v_A(z)$. Letting $z \rightarrow -\infty$ gives $A(0) = 0$, and (P 6.3) becomes $v_{\iota A} = v_A$. If p is odd, then $-v_A(-z) = v_A(z) + 2P(0)$, and (P 6.3) becomes $v_{\iota A}(z) = v_A(z) + 2P(0) + s_p A(0)$. Letting $z \rightarrow -\infty$ shows that the constant vanishes. In both cases $v_{\iota A} = v_A$, hence $\iota A = A$ by the uniqueness of the form (P 3.1).

(ii). On the subspace $\{\alpha_j = 0, j \text{ odd}\}$ we have $\iota A = A$, and (P 6.2) shows that R_A is an odd polynomial. The equations $R_r = 0$ with r even therefore hold identically there. With $m = \lfloor d/2 \rfloor$,

there remain the m equations $R_1, R_3, \dots, R_{2m-1}$ in the m unknowns $\alpha_2, \alpha_4, \dots, \alpha_{2m}$. By (2.1) their weights are

$$\begin{array}{l|l} \text{variables} & 2, 4, \dots, 2m \\ \text{equations} & 2d, 2d-2, \dots, 2(d-m+1). \end{array}$$

Their highest weighted parts are the restrictions of those of the full system. The highest weighted parts of the omitted equations vanish identically on the subspace, because H_A is an odd polynomial when $\iota A = A$. A common zero of the restricted leading forms is therefore a polynomial A with $A \mid H_A$, and (R_d) gives $A = \rho^d$. The finite-cover argument of Theorem 2.1, with $\alpha_{2i} = y_i^{2i}$, gives

$$\text{length } \mathcal{X}_p^{\text{sym}} = \prod_{i=1}^m \frac{2(d-i+1)}{2i} = \binom{d}{m},$$

and $\binom{d}{m} = F_p$. Without (R_d) , the same substitution and Bézout's inequality for isolated solutions, used as in the first part of the proof of Theorem 2.1, bound the sum of the multiplicities of the isolated points of $\mathcal{X}_p^{\text{sym}}$ by the same quotient. This holds for every p .

(iii). A closed subscheme of a finite reduced scheme is reduced, as in the proof of Proposition 3.1, and a reduced point of a square system is a simple solution of it. The map (P 6.1) negates the roots of $\bar{A}$, so it acts on the labels by $L \mapsto -L$, and the labelling is a bijection.

(iv). For even p the polynomial A is odd, so $A(0) = 0$. Let p be odd, and suppose that a point of $\mathcal{X}_p^{\text{sym}}$ is a pulse, $A(0) = 0$. Then $P_A(0) = 0$ by (4.2). If $\mathcal{X}_p$ is reduced, this contradicts Proposition 3.1. If A has real coefficients, then v_A is real on the real axis, and it tends to zero with all its derivatives at both ends. Integrating (P 1.1) along the real axis, and using $P_A(0) = 0$, gives $\int v_A^2 dz = 0$. Hence $v_A = 0$, which is impossible. $\square$

5 Elliptic pairs

Theorem 5.1 (Fixed-lattice elliptic enumeration). *Assume (R_{p-1}) . For every fixed nonsingular lattice,*

$$\text{length } \mathcal{E}_p(g_2, g_3) = M_p := 2 \binom{2p-2}{p-2}. \quad (5.1)$$

If, in addition, $\mathcal{X}_p$ is reduced, then $\mathcal{E}_p(g_2, g_3)$ is reduced for (g_2, g_3) in a nonempty Zariski-open subset of $\{g_2^3 - 27g_3^2 \neq 0\}$. In particular, if $2p-1$ is prime, a generic fixed lattice admits exactly M_p normalized elliptic operator-profile pairs, all simple.

Thus the elliptic assertion on an arbitrary lattice is a count with multiplicities. Exceptional lattices, on which distinct pairs merge, do occur and cannot be discarded when counting distinct solutions; Section P 8.2 determines them at order three.

The proof has five steps. We eliminate the operator, which leaves $p-1$ equations in the $p-1$ profile unknowns. We assign weights to the unknowns, under which the highest weighted parts of the equations are obtained by setting $g_2 = g_3 = 0$. We show that at this cusp the equations have only the trivial common zero; this is where (R_{p-1}) enters, and it means that there are no solutions at weighted infinity for any g_2, g_3 . Weighted Bézout then gives the length M_p on every lattice. Finally, at a nodal cubic the elliptic problem becomes the equal-endpoint case of the rational-exponential problem, whose solutions are simple by Proposition 3.1 when $\mathcal{X}_p$ is reduced, and simplicity spreads from the nodal parameter value to nearby nonsingular parameters.

Proof. Step 1: eliminating the operator. The coefficient at pole order $2p$ vanishes by the choice of s_p . The coefficients at orders $2p-1, \dots, p$ determine $a_{p-1}, \dots, a_0$ successively. At each step the coefficient of the new a_j is the nonzero rational leading coefficient of $D^j V$. This is a polynomial elimination, valid over any $\mathbb{Q}$ -algebra. The residual has pole order at most $p-1$ and zero residue. It lies in the span of

$$\{1\} \cup \{\mathcal{D}^j X : 0 \leq j \leq p-3\}.$$

Hence $p-1$ equations remain in the $p-1$ profile coefficients u_j, b .

Step 2: weights. Give D, X, g_2, g_3 weights $1, 2, 4, 6$, respectively, and give the profile weight p . Then u_j has weight $p-j-2$, and b has weight p . After the triangular elimination, the residual principal-part coefficients and the residual Laurent constant have weights

$$\begin{array}{c|c} \text{variables} & 1, 2, \dots, p-2, p \\ \text{equations} & p+1, p+2, \dots, 2p-2, 2p. \end{array} \quad (5.2)$$

The Laurent expansions and the eliminations are polynomial in g_2, g_3 , so these statements extend to the singular cubics. For fixed g_2, g_3 , the highest weighted parts of the equations in the profile variables are obtained by setting $g_2 = g_3 = 0$, since g_2 and g_3 carry positive weight.

Step 3: the cusp. At the cusp $g_2 = g_3 = 0$, $X = z^{-2}$ and $Y = -2z^{-3}$, so the profile is a rational function. Write $V = b + f(z)$, where f has its sole pole at zero and vanishes at infinity. The constant and decaying parts of the equation give

$$a_0 b + \frac{1}{2} b^2 = 0, \quad (P+b)(D)f + \frac{1}{2} f^2 = 0.$$

The leading term of f at infinity forces $a_0 + b = 0$. It follows that $b = a_0 = 0$. Under the polynomial Laplace correspondence

$$\sum_k A_k \rho^k \mapsto \sum_k k! A_k z^{-k-1},$$

differentiation corresponds to multiplication by $-\rho$, and multiplication of the rational functions corresponds to continuous convolution. After a nonzero scalar normalization, the equation for f therefore gives $A \mid H_A$ for a monic A of degree $p-1$. By (R_{p-1}) , $A = \rho^{p-1}$. Thus $f = c_* z^{-p}$, and all the variables in (P 4.2) vanish. The leading equations have no nonzero common zero.

Step 4: the length. The same finite-cover Bézout argument as in Theorem 2.1, now with the weights (5.2), gives

$$\text{length } \mathcal{E}_p(g_2, g_3) = \frac{(p+1)(p+2) \cdots (2p-2)(2p)}{(p-2)! p} = 2 \binom{2p-2}{p-2}.$$

For $p=3$, this is $(4 \cdot 6)/(1 \cdot 3) = 8$. Empty products are one, so the calculation includes $p=2$. It proves this length at every (g_2, g_3) , including the singular parameters used below; at the cusp itself the whole length sits at the single solution where every unknown vanishes. It also makes the whole coefficient family finite over $\mathbb{C}[g_2, g_3]$: sufficiently large weighted monomials reduce to bounded weight using the leading equations, which are independent of g_2, g_3 . This finiteness is what allows simplicity to be transferred from the nodal parameter value to nearby nonsingular parameters in the last step.

Step 5: the node. To prove generic simplicity, consider the nodal parameters

$$g_2 = \frac{1}{12}, \quad g_3 = -\frac{1}{216}. \quad (5.3)$$

In the logistic coordinate they are realized by

$$X = Q^2 - Q + \frac{1}{12}, \quad Y = (2Q - 1)Q(1 - Q), \quad DQ = Q(1 - Q).$$

The nodal coordinate ring is

$$\mathbb{C}[X, Y] = \{f \in \mathbb{C}[Q] : f(0) = f(1)\}.$$

Indeed the right side is $\mathbb{C} + Q(1 - Q)\mathbb{C}[Q]$; it is generated by X, Y , since, with $S = Q(1 - Q)$, $S = 1/12 - X$, $SQ = (Y + S)/2$, and $Q^2 = Q - S$. The pole filtration is the ordinary polynomial degree in Q . Consequently the nodal specialization of (P 4.2) has the unique form

$$V = \beta + v_A, \quad A \text{ monic of degree } p - 1, \quad A(0) = 0,$$

where β is the common endpoint. The coefficient change is invertible and affine over $\mathbb{Q}$.

The matching equations become, as equations of schemes,

$$A \in \mathcal{Z}_p, \quad P = P_A - \beta, \quad \beta(\beta - 2P_A(0)) = 0. \quad (5.4)$$

To see this directly, the constant equation is $P(0)\beta + \beta^2/2 = 0$. The remaining polynomial identity is $(P + \beta)(D)v_A + v_A^2/2 = 0$, equivalent by Proposition P 3.1 to $A \mid S_A$ and $P + \beta = P_A$. The coefficient changes and monic division used here work over arbitrary $\mathbb{C}$ -algebras, so the correspondence preserves local multiplicities.

If $\mathcal{X}_p$ is reduced, Proposition 3.1 gives S_p simple points of $\mathcal{Z}_p$, with $P_A(0) \neq 0$ at every point. Each quadratic in (5.4) then has two distinct roots. The nodal fiber has $2S_p = M_p$ simple points. In the finite coefficient family, the image of the closed locus where the matching Jacobian vanishes is closed. Its complement contains (5.3), and hence meets the nonsingular-lattice locus in a nonempty open set. Every fiber in that open set is reduced, proving the theorem. $\square$

The factor two in M_p retains the two choices of background in (P 2.5). That transformation preserves the evolution equation and the lattice; it changes the operator P and the speed. At the nodal fiber it exchanges the two roots in (5.4), which are distinct when $\mathcal{X}_p$ is reduced. In that case the two backgrounds are distinct on a generic lattice, and the M_p pairs represent $M_p/2$ elliptic waves, up to translations and Galilean shifts, each counted with both of its backgrounds. On special nonsingular lattices the two backgrounds may coalesce, as Section P 8.2 shows.

Corollary 5.2. *Let p be even and assume (R_{p-1}) . On every fixed nonsingular lattice, the elliptic pairs whose operator is an even polynomial form a closed subscheme $\mathcal{E}_p^{\text{sym}}(g_2, g_3)$ of $\mathcal{E}_p(g_2, g_3)$, defined by $u_j = 0$ for odd j , and*

$$\text{length } \mathcal{E}_p^{\text{sym}}(g_2, g_3) = 2F_p. \quad (5.5)$$

If $\mathcal{X}_p$ is reduced, $\mathcal{E}_p^{\text{sym}}(g_2, g_3)$ is reduced for (g_2, g_3) in a nonempty Zariski-open set of nonsingular parameters.

Proof. An even operator has an even profile, by Corollary P 5.3. Conversely, if the profile (P 4.2) is even, the odd part of (P 1.1) is $\sum_{j \text{ odd}} a_j D^j V = 0$; the functions $D^j V$ have distinct pole orders, so $a_j = 0$ for odd j . On the subspace $u_j = 0$, j odd, the residual of Step 1 is even, and there remain the equations and unknowns of even weight in (5.2):

$$\begin{array}{c|c} \text{variables} & 2, 4, \dots, p-2, p \\ \text{equations} & p+2, p+4, \dots, 2p-2, 2p. \end{array}$$

The omitted leading forms vanish identically on the subspace, so the remaining ones have only the trivial common zero, by Step 3. Weighted Bézout gives

$$\frac{(p+2)(p+4)\cdots(2p-2)(2p)}{2\cdot 4\cdots(p-2)p} = 2\binom{p-1}{p/2-1} = 2F_p.$$

At the node, reflection is $Q \mapsto 1 - Q$. An even nodal profile is $\beta + v_A$ with $\iota A = A$, so the nodal fiber is (5.4) with $A \in \mathcal{X}_p^{\text{sym}}$. If $\mathcal{X}_p$ is reduced, it has $2F_p$ simple points, by Theorem 4.1 and Proposition 3.1, and Step 5 applies without change. $\square$

6 What is known at each order

Each formula needs two facts at a given p . The first is the rigidity condition (R_{p-1}) of Section P 3. It gives N_p , M_p , F_p and $2F_p$ as counts with multiplicity: the multiplicities of the pairs add up to these numbers. The second is that every rational-exponential pair is simple, that is, $\mathcal{X}_p$ is reduced. Then the pairs are distinct, and the elliptic pairs are distinct on a generic lattice. On a special lattice several elliptic pairs can merge, while the count with multiplicity stays the same; Section P 8.2 shows this at order three. Table 1 records what is known about each fact.

Table 1: What is known. For elliptic pairs, distinctness is asserted only on a generic lattice.

Statement	Known for
Counts with multiplicity	$2p - 1$ a prime or a prime power (proved); every $p \leq 73$ (exact certificate); every $p \leq 573$ (finite-field computations)
All N_p pairs distinct	$2p - 1$ prime (proved); $p = 5, 8, 11$ (exact computations); hence every $p \leq 12$
All F_p pairs of the pure subfamily distinct	whenever all N_p pairs are distinct; every $p \leq 14$ (exact computations)
Both statements, every p	conjectured

The prime-index argument excluded solutions at infinity by reducing the continuous convolution modulo ℓ . The same argument proves (2.4) whenever $2p - 1 = \ell^e$ is an odd prime power (Section 7). It gives the count with multiplicity, but it does not show that the pairs are distinct.

More generally, the exact arithmetic certificates supplied with the paper establish (R_d) for every $d \leq 72$, hence

$$\text{length } \mathcal{X}_p = \binom{2p-1}{p-1}, \quad \text{length } \mathcal{E}_p(g_2, g_3) = 2\binom{2p-2}{p-2} \quad (2 \leq p \leq 73). \quad (6.1)$$

In the same range, Theorem 4.1 and Corollary 5.2 give $\text{length } \mathcal{X}_p^{\text{sym}} = F_p$ and, for even p , $\text{length } \mathcal{E}_p^{\text{sym}}(g_2, g_3) = 2F_p$. Together with the ancillary proof, the certificates also cover infinite families of higher orders. Finite-field Gröbner computations, checked by independent pipelines, establish (R_d) for every $d \leq 572$; they extend (6.1) and the two symmetric lengths, as counts with multiplicity, to every $2 \leq p \leq 573$ (repository, directory `certificates/rigidity_572/`). The certificates hold for all complex coefficients, not only real or rational ones. The ancillary proof and the full certificate are described in Appendix P A.

Equation (6.1) is a statement about counts with multiplicity only. At composite indices, exact computations show that all pairs are simple at $p = 5$, $p = 8$ and $p = 11$ (Appendix P A). With Theorem P 3.4, all pairs are therefore distinct for every $p \leq 12$, and Theorem 5.1 gives M_p distinct elliptic pairs on a generic lattice at these orders. Inside the two pure subfamilies the computation is much smaller, and it shows that the F_p pairs are distinct for every $p \leq 14$, which includes the composite indices 9, 15, 21, 25 and 27.

7 Rigidity at a prime-power index

Let $2p - 1 = \ell^e$ be an odd prime power. For $\deg A = (\ell^e - 1)/2$, the normalized continuous convolution satisfies

$$\ell^e H_A(\rho) \equiv u \rho^{\ell^e} \pmod{\ell}, \quad u \neq 0.$$

Every bilinear coefficient other than the leading one has denominator valuation at most $e - 1$, whereas the leading coefficient has valuation $-e$. This follows either from the factorial formula or from

$$\frac{i!j!}{(i+j+1)!} = \sum_{r=0}^j \frac{(-1)^r \binom{j}{r}}{i+r+1}.$$

Lemma 8.1(a) below, applied to the coefficients of $\text{rem}_A(\ell^e H_A)$, then gives (R_{p-1}) , in the same way as in Step 3 of the proof of Theorem P 3.4. This argument controls only the leading system, so it gives no simple-root lifting statement for the discrete matching system.

The computations referred to above are described in Appendix P A of the paper and in the repository <https://github.com/migita/binomial-count-meromorphic-waves>, directories `certificates/` and `paper/anc/`.

8 Two algebraic arguments from an earlier version, and the proof of the conservation lemma

The paper now proves Step 3 of Theorem P 3.4 by rescaling with a root of A , and the conservation of the number of elliptic pairs by a deformation argument (Lemma P 4.2). An earlier version used the two algebraic arguments of this section instead. The first of them is also what Section 7 uses. The paper gives only a sketch of the proof of Lemma P 4.2; Section 8.2 contains the proof in full.

8.1 Reduction modulo ℓ

Part (a) of the lemma transfers the absence of solutions at infinity from characteristic ℓ to characteristic zero; part (b) says that the solutions themselves can be reduced modulo ℓ . Let $\mathbb{Z}_{(\ell)}$ be the ring of rational numbers whose denominator is prime to ℓ . For a polynomial G in $\alpha_1, \dots, \alpha_d$, write G^{top} for its part of highest weight.

Lemma 8.1 (Reduction modulo ℓ). *Let $G_1, \dots, G_m \in \mathbb{Z}_{(\ell)}[\alpha_1, \dots, \alpha_d]$, and suppose that the reductions modulo ℓ of $G_1^{\text{top}}, \dots, G_m^{\text{top}}$ have no common zero over $\overline{\mathbb{F}}_\ell$ other than the origin. Then:*

- (a) $G_1^{\text{top}}, \dots, G_m^{\text{top}}$ have no common zero over $\mathbb{C}$ other than the origin;
- (b) every common zero of $G_1, \dots, G_m$ with coordinates in an algebraic closure K of $\mathbb{Q}_\ell$ is integral: its coordinates lie in the valuation ring of K .

Proof. Let v be the valuation of K with $v(\ell) = 1$, let $\mathcal{O}$ be its valuation ring, and recall that the residue field of $\mathcal{O}$ is $\overline{\mathbb{F}}_\ell$ and that, K being algebraically closed, every rational number is the valuation of some element of K .

For (b), let $\alpha \in K^d$ be a common zero of the G_i , put $t = \min_j v(\alpha_j)/j$, and suppose that $t < 0$. Choose $\lambda \in K$ with $v(\lambda) = t$, and put $\beta_j = \alpha_j/\lambda^j$. Then every β_j lies in $\mathcal{O}$, and at least one of them is a unit. Decompose $G_i = \sum_{w \leq W_i} G_i^{(w)}$ into its parts of weight w , so that $G_i^{(W_i)} = G_i^{\text{top}}$. A part of weight w satisfies $G_i^{(w)}(\alpha) = \lambda^w G_i^{(w)}(\beta)$. Dividing the equation $G_i(\alpha) = 0$ by λ^{W_i} gives

$$G_i^{\text{top}}(\beta) = - \sum_{w < W_i} \lambda^{w-W_i} G_i^{(w)}(\beta).$$

Each term on the right has positive valuation, because $v(\lambda^{w-W_i}) = (w-W_i)t > 0$ and $G_i^{(w)}(\beta) \in \mathcal{O}$. Reducing modulo the maximal ideal of $\mathcal{O}$, we find that the reduction of β , which is not the origin, is a common zero of the reduced polynomials G_i^{top} . This contradicts the hypothesis. Hence $t \geq 0$, which is (b).

For (a), suppose that the G_i^{top} have a common complex zero other than the origin. They are weighted-homogeneous with rational coefficients, so after fixing one nonzero coordinate to be 1 the weak Nullstellensatz gives such a zero α with coordinates in $\overline{\mathbb{Q}} \subset K$. Define t , λ and β as before, now for either sign of t . Then $G_i^{\text{top}}(\beta) = \lambda^{-W_i} G_i^{\text{top}}(\alpha) = 0$, and the reduction of β is again a forbidden common zero. $\square$

The argument needs the algebraic closure of the residue field; a search for zeros over $\mathbb{F}_\ell$ alone would not suffice.

In that version, Step 3 of the proof of Theorem P 3.4 applied the lemma to $G_r = \ell R_r$, $0 \leq r < d$, with $\ell = 2p - 1$ prime. By Section 2, G_r^{top} is the coefficient of ρ^r in $\text{rem}_A(\ell H_A)$, and $\ell H_A(\rho) \equiv (-1)^d \rho^\ell \pmod{\ell}$; this is the case $e = 1$ of the congruence in Section 7. So the reduced polynomials G_r^{top} vanish together exactly when $\overline{A} \mid \rho^\ell$, that is, when $\overline{A} = \rho^d$, which is the origin $\alpha_1 = \dots = \alpha_d = 0$. The hypothesis of the lemma holds. Part (a) of the lemma says that $\text{rem}_A H_A = 0$ only for $A = \rho^d$, which is (R_d) . Part (b) says that every solution of the matching equations with coordinates in K is integral.

8.2 Proof of Lemma P 4.2

Lemma P 4.2 concerns polynomials $E_1, \dots, E_n$ in unknowns $x = (x_1, \dots, x_n)$ and parameters $t = (t_1, \dots, t_m)$, weighted-homogeneous for positive integer weights of the x_j and the t_k , whose leading system $E_i(x; 0) = 0$, $1 \leq i \leq n$, has no solution in $\mathbb{C}^n$ other than $x = 0$; it states that (i) for every t the system $E(x; t) = 0$ has finitely many solutions, which stay bounded while t stays bounded, (ii) their number, counted with multiplicity, is the same for every $t \in \mathbb{C}^m$, and (iii) the t for which some solution is not simple form an algebraic subset of $\mathbb{C}^m$. Since the t_k have positive weight, $E_i(x; 0)$ is the part of E_i of highest weight in the unknowns alone.

Proof. Since the leading system has only the zero solution, the polynomials in x modulo the combinations $\sum_i h_i E_i(x; 0)$ form a finite-dimensional space [3, Chapter 5, §3]. It is spanned by finitely many monomials $m_1 = 1, m_2, \dots, m_r$. We claim that every polynomial f in (x, t) can be written

$$f = \sum_{i=1}^r c_i(t) m_i + \sum_{i=1}^n h_i E_i \tag{8.1}$$

with polynomials c_i in t and h_i in (x, t) . It is enough to prove this for weighted-homogeneous f , by induction on the weight w of f ; a polynomial of weight zero is a constant. Write $f = f(x, 0) + \sum_k t_k f_k$ and $E_i = E_i(x; 0) + \sum_k t_k e_{ik}$, with weighted-homogeneous f_k and e_{ik} , and write $f(x, 0) = \sum_i c_i m_i + \sum_i h_i E_i(x; 0)$ with constants c_i and with polynomials h_i in x . Taking the parts of weight w on both sides, we may assume that $c_i = 0$ unless m_i has weight w , and that every h_i is weighted-homogeneous, of weight w minus the weight of E_i . Then

$$f = \sum_i c_i m_i + \sum_i h_i E_i + \sum_k t_k \left(f_k - \sum_i h_i e_{ik} \right),$$

and each bracket is weighted-homogeneous of weight less than w , because t_k has positive weight. The induction hypothesis applies to the brackets, and this proves (8.1).

Apply (8.1) to the products $x_j m_i$: modulo the equations, that is, modulo the combinations $\sum_i h_i E_i$, we have $x_j m_i = \sum_l M_{il}^{(j)}(t) m_l$ with a matrix $M^{(j)}$ of polynomials in t . Put $\chi_j(x_j; t) = \det(x_j - M^{(j)}(t))$. Multiplying the vector identity by the adjugate of the matrix $x_j - M^{(j)}(t)$ shows that $\chi_j m_i$ is a combination of the E_i for every i (the Cayley–Hamilton argument); for $m_1 = 1$ this is χ_j itself. Thus at every solution x_j is a root of $\chi_j(\cdot; t)$, a monic polynomial of degree r whose coefficients are polynomials in t . There are therefore at most r^n solutions, and they stay bounded while t stays bounded, because the roots of a monic polynomial are bounded in absolute value by one plus the largest absolute value of its coefficients. This proves (i).

For (ii), let $t(s)$, $0 \leq s \leq 1$, be the segment between two parameter values. By (i) there is a ball $|x| < \varrho$ that contains all solutions of $E(x; t(s)) = 0$ for all s . The maps $x \mapsto E(x; t(s))$ then form a continuous family of holomorphic maps, each with finitely many zeros in the ball and none on the sphere $|x| = \varrho$. For such a family the number of zeros in the ball, counted with multiplicity, does not depend on s : it is the degree of the map $E/|E|$ from that sphere to the unit sphere, which is invariant under deformation. This is Rouché’s theorem in several variables [5, Appendix B], [1, Sections 5.2 and 5.4]. The multiplicity of a zero in this statement is the local degree of the map. It equals the number of nearby solutions of a generic small perturbation of the system, and the dimension of the local algebra [1, Sections 5.2 and 5.4], so it is the multiplicity of Section P 3. Any two parameter values are joined by a segment, which proves (ii).

For (iii), the parameters in question are the projection to $\mathbb{C}^m$ of the algebraic set defined in $\mathbb{C}^n \times \mathbb{C}^m$ by $E_1 = \cdots = E_n = 0$ and $\det(\partial E_i / \partial x_j) = 0$. The monic polynomials χ_j are combinations of the E_i . Eliminate the unknowns one at a time: at each step the polynomial χ_j for the unknown being eliminated has leading coefficient 1, so by the Extension Theorem of elimination theory [3, Chapter 3, §1] every solution of the eliminated equations extends. The projection is therefore the common zero set of the polynomials in t alone that are combinations of the E_i and the Jacobian determinant; finitely many of them suffice, by Hilbert’s basis theorem. $\square$

The boundedness in (i) can also be seen directly, by the rescaling that the paper describes before the lemma. Let w_j , ω_k and W_i be the weights of x_j , t_k and E_i , and put $N(x) = \max_j |x_j|^{1/w_j}$. Suppose that there were solutions $(x^{(\nu)}, t^{(\nu)})$ with $t^{(\nu)}$ bounded and $\lambda_\nu = N(x^{(\nu)}) \rightarrow \infty$. Put $y_j^{(\nu)} = \lambda_\nu^{-w_j} x_j^{(\nu)}$ and $\tau_k^{(\nu)} = \lambda_\nu^{-\omega_k} t_k^{(\nu)}$. Then $N(y^{(\nu)}) = 1$, $\tau^{(\nu)} \rightarrow 0$, and $E_i(y^{(\nu)}; \tau^{(\nu)}) = \lambda_\nu^{-W_i} E_i(x^{(\nu)}; t^{(\nu)}) = 0$ by weighted homogeneity. The set $N(y) = 1$ is compact, so the $y^{(\nu)}$ have a limit point y^* . By continuity it satisfies $N(y^*) = 1$ and $E_i(y^*; 0) = 0$, and is thus a nonzero solution of the leading system, contrary to the hypothesis. This is the proof of Lemma 8.1(b) with the ordinary absolute value in place of the ℓ -adic one; compare Step 3 in the proof of Theorem P 3.4.

8.3 Conservation of number by flatness

Lemma P 4.2 of the paper concerns n polynomial equations in n unknowns and in parameters t , weighted-homogeneous for positive weights of the unknowns and of the parameters, whose leading system, the one at $t = 0$, has only the zero solution. Its parts (i) and (ii) say that every system of the family has finitely many solutions, which stay bounded while t stays bounded, and that their number, counted with multiplicity, does not depend on t . Section 8.2 proves this by a deformation argument. What follows is a second, algebraic proof. It is Step 4 of the elliptic count in that version, and it is written for the family used there: the residual equations in the proof of Theorem 5.1, to which the step numbers below refer, with $t = (g_2, g_3)$. The general case is the same, with $\mathbb{C}[t]$ in place of $\mathbb{C}[g_2, g_3]$ and the number of parameters in place of two.

Let $E_1, \dots, E_{p-1}$ be the residual equations of Step 1. They are polynomials in the $p-1$ profile unknowns and in g_2, g_3 , weighted-homogeneous for the weights of Step 2, and

$$\mathcal{A} = \mathbb{C}[g_2, g_3][u_0, \dots, u_{p-3}, b]/(E_1, \dots, E_{p-1})$$

is the coordinate ring of the family of all the systems $\mathcal{E}_p(g_2, g_3)$. By Step 3 the cusp system has the origin as its only solution, so $\mathcal{A}/(g_2, g_3)\mathcal{A}$ is a finite-dimensional vector space. The graded form of Nakayama's lemma then shows that $\mathcal{A}$ is a finitely generated module over $\mathbb{C}[g_2, g_3]$: homogeneous elements whose classes span the cusp ring generate it. In particular every fiber is finite, and every unknown satisfies a monic equation with coefficients in $\mathbb{C}[g_2, g_3]$, so the solutions stay bounded while (g_2, g_3) stays bounded. This proves (i) and makes precise the statement that solutions do not escape to infinity on any lattice.

The family is also flat. It is defined by $p-1$ equations in $p+1$ variables, so each of its components has dimension at least two, and it is finite over the plane, so its dimension is exactly two. Hence $E_1, \dots, E_{p-1}$ is a regular sequence and $\mathcal{A}$ is Cohen–Macaulay. A Cohen–Macaulay algebra that is finite over a regular ring of the same dimension is flat over it [4, Theorem 23.1], and a finite flat module over $\mathbb{C}[g_2, g_3]$ is locally free. Consequently the number of solutions of $\mathcal{E}_p(g_2, g_3)$, counted with multiplicity, is the same at every point of the (g_2, g_3) -plane, the singular cubics included. At the cusp itself all of this multiplicity sits at the single solution where every unknown vanishes. This proves (ii).

The paper finds the common number at the node, in its Step 5. In this note it is the Bézout number of Step 4 in the proof of Theorem 5.1.

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